

DENDROIDAL SPACES

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The main goal of these notes is to review the definition of complete dendroidal Segal spaces as a model of ∞ -operads. The topic is divided in the notes for two talks. These notes correspond to the first one, where we study Reedy model structures and the category of dendroidal spaces. In the next session, we review left Bousfield localization, dendroidal Segal spaces and complete dendroidal Segal spaces. The original reference for this topic is [2], but we followed a more modern write up in [3].

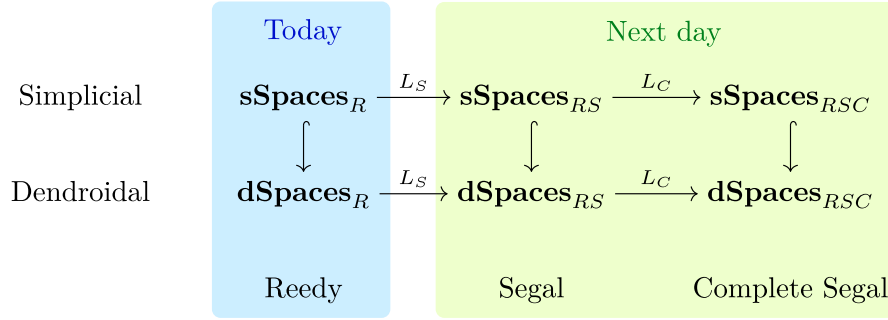


FIGURE 1. Organization of the talks related to these notes

Let A be a small category, C be a model category. The category of $[A, C]$ has two natural model structures:

- The *projective model structure*, denoted $[A, C]_{proj}$, where:
 - a weak equivalence is a natural transformation $f : F \rightarrow G$ such that $f_a : F(a) \rightarrow G(a)$ are weak equivalences of C for all $a \in A$,
 - a fibration is a natural transformation $f : F \rightarrow G$ such that $f_a : F(a) \rightarrow G(a)$ are fibrations of C for all $a \in A$,
 - a cofibration is a natural transformation with the left lifting property against trivial fibrations,
 - and a fibrant object is an object F such that $F(a)$ is fibrant for any $a \in A$.
- The *injective model structure*, denoted $[A, C]_{inj}$, where:
 - a weak equivalence is a natural transformation $f : F \rightarrow G$ such that $f_a : F(a) \rightarrow G(a)$ are weak equivalences of C for all $a \in A$,
 - a cofibration is a natural transformation $f : F \rightarrow G$ such that $f_a : F(a) \rightarrow G(a)$ are cofibrations of C for all $a \in A$,
 - a fibration is a natural transformation with the right lifting property against trivial cofibrations,
 - and a cofibrant object is an object F such that $F(a)$ is cofibrant for any $a \in A$.

Although these model structures may not exist, the existence can be proven by imposing some conditions on C . If C is cofibrantly generated, then the projective model structure exists. If C is combinatorial, then both model structures exists.

Unfortunately, neither of these two model structures has the properties that we would want for the model structure over **dSpaces**: on the projective model structure, there is no explicit characterization of cofibrations and cofibrant objects, and dually, in the injective model structure the same occurs with the fibrations and fibrant objects. In section 1, we introduce an “intermediate” model structure: the Reedy model structure [1, 3]. This construction requires the indexing category A to be Reedy, but has the same weak equivalences as the previous two model structures together with explicit fibrations and cofibrations. Later in section 2, we introduce the category of dendroidal spaces and study the Reedy model structure applied to this particular example.

1. REEDY CATEGORIES

Definition 1.1. A category R is *Reedy* if it has

- a *degree* function $d : \text{Obj } R \rightarrow \mathbb{N}$,
- a set of *positive* morphisms $R^+ \subseteq \text{Mor } R$,
- and a set of *negative* morphisms $R^- \subseteq \text{Mor } R$;

such that for all morphism $f : r \rightarrow s$

- (i) R^+ and R^- are closed under composition,
- (ii) f can be factored as $f = p \circ n$, where p is positive and n is negative, and this factorization is unique up to isomorphism,
- (iii) if $f \in R^+$ (resp. $f \in R^-$), then $d(r) \leq d(s)$ (resp. $d(r) \geq d(s)$),
- (iv) f is an isomorphism $\iff d(r) = d(s)$,
- (v) if $f \in R^+$ (resp. $f \in R^-$), Θ is an automorphism of r (resp. s), and $f\Theta = f$ (resp. $\Theta f = f$), then $\Theta = \text{id}$.

If a morphism is positive (resp. negative) and not an isomorphism, then we call it *strictly positive* (resp. *strictly negative*) and denote it by $f : r \rightarrow^+ s$ (resp. $f : r \rightarrow^- s$). In the literature, the previous definition is usually called generalized Reedy category, leaving the name of Reedy category to a category with the conditions from (i) to (iv) and the factorization of (ii) being strictly unique. In these notes, we have decided to stick to the notation of [3], calling Reedy to a generalized Reedy category, and *strict Reedy* to a Reedy category. Furthermore, a Reedy category is strict Reedy if and only if it does not have nontrivial isomorphisms.

Example 1.2. (1) Δ is strict Reedy with $d([n]) = n$, the injections as the positive morphisms, and the surjections as the negative ones.
(2) Ω is Reedy with $d(T)$ being the number of vertices of T , the composition of faces and isomorphisms as the positive morphisms, and the composition of degeneracies and isomorphism as the negatives ones.

Let C be a model category. Consider the category $[R^{\text{op}}, C]$ of C -valued presheaves on R . Because C is complete and cocomplete, for any $r \in R$ and $X : R^{\text{op}} \rightarrow C$, there exist

- the r -th *matching object* of X

$$M_r X = \lim_{s \rightarrow^+ r} X(s),$$

- and the r -th *latching object* of X

$$L_r X = \text{colim}_{r \rightarrow^- s} X(s).$$

In addition, there is a natural $\text{Aut}(r)$ -equivariant maps $L_r X \rightarrow X(r) \rightarrow M_r X$. As a consequence, for any morphism $f : X \rightarrow Y$ in $[R^{\text{op}}, C]$, there exists

- the r -th *matching map* of f

$$X(r) \rightarrow M_r X \times_{M_r Y} Y(r),$$

- and the r -th *latching map* of f

$$L_r Y \sqcup_{L_r X} X(r) \rightarrow Y(r).$$

Example 1.3. If C is $\mathbf{sSet}_Q$ the category of simplicial sets with the Kan-Quillen model structure and $R = \Delta$, then $\mathbf{sSpaces} = [\Delta^{\text{op}}, \mathbf{sSet}_Q]$ is the category of simplicial spaces. In this case, for all $[n] \in \Delta$ and $X : \Delta^{\text{op}} \rightarrow \mathbf{sSet}$:

- the latching object $L_{[n]}X$ is the union of all degenerate n -cells,
- and the matching object $M_{[n]}X$ is the powering $X^{\partial\Delta[n]}$ of the boundary of the n -simplex into X , which is equivalent to the n -cells of the $(n-1)$ -skeleton of X .

In section 2 we study how to characterize latching and matching objects in the case of $R = \Omega$ and $C = \mathbf{sSet}_Q$, i.e., studying the category of dendroidal space $\mathbf{dSpaces} = [\Omega^{\text{op}}, \mathbf{sSet}_Q]$. Let $\mathbf{BAut}(r)$ be the category with one object and homset equal to the group $\text{Aut}(r)$. The Reedy model structure exists for suitable conditions on C :

Theorem 1.4 ([3, Theorem 10.14]). *Let R be a Reedy category, and C be a cofibrantly generated model category. Then there exists a cofibrantly generated model structure on $[R^{\text{op}}, C]$, called the Reedy model structure and denoted $[R^{\text{op}}, C]_R$, where a map $f : X \rightarrow Y$ is:*

- a weak equivalence if $f_r : X(r) \rightarrow Y(r)$ are weak equivalences of C for all $r \in R$,
- a fibration if the r -th matching map is a fibration of C for all $r \in R$,
- or a cofibration if the r -th latching map is a cofibration of $[\mathbf{BAut}(r)^{\text{op}}, C]_{\text{proj}}$ for all $r \in R$.

In addition, an object $X : R^{\text{op}} \rightarrow C$ is:

- fibrant $\iff X(r) \rightarrow M_r X$ is a fibration of C for all $r \in R$,
- or cofibrant $\iff L_r X \rightarrow X(r)$ is a cofibration of $[\mathbf{BAut}(r)^{\text{op}}, C]_{\text{proj}}$ for all $r \in R$.

It can be seen that every fibration in $[R^{\text{op}}, C]_R$ is a fibration in $[R^{\text{op}}, C]_{\text{proj}}$. Therefore, the Reedy model structure has the weak equivalences and some fibrations of the projective model structure. In fact, the identity functors induce a Quillen equivalence between the projective and Reedy model structures. Additionally, if C is combinatorial, then $[R^{\text{op}}, C]_R$ is also a combinatorial model category. In this case, the identity functors also induce a Quillen equivalence between the injective and Reedy model structures

$$[R^{\text{op}}, C]_{\text{proj}} \simeq_Q [R^{\text{op}}, C]_R \simeq_Q [R^{\text{op}}, C]_{\text{inj}}.$$

In the case of R being a strict Reedy category, like for example with $R = \Delta$, the characterization of the cofibrations in the Reedy model structure simplifies to only involve the model structure on C :

Proposition 1.5 ([3, Proposition 10.15]). *If R has no nontrivial automorphisms, then a map $f : X \rightarrow Y$ is cofibrant in $[R^{\text{op}}, C]_R$ if and only if the r -th latching map is a cofibration of C for all $r \in R$.*

Remark 1.6. The model structure of Theorem 1.4 is not the only option: if C is combinatorial, one could also define another one by replacing the cofibrations by the maps for which the r -th matching maps are cofibrations of C for all $r \in R$, and fibrations by the maps for which the r -th latching maps are fibrations in $[\mathbf{BAut}(r)^{\text{op}}, C]_{\text{inj}}$ for all $r \in R$. In this case, we obtain an “injective” Reedy model structure, which is also Quillen equivalent to the other model structures over $[R^{\text{op}}, C]$.

2. DENDROIDAL SPACES

In this section we study the category $\mathbf{dSpaces} = [\Omega^{\text{op}}, \mathbf{sSet}_Q]$ of dendroidal spaces. Any dendroidal set A can be viewed as a *discrete dendroidal space* $\text{dis } A \in \mathbf{dSpaces}$ defined as, for any $T \in \Omega$ and $n \in \mathbb{N}$,

$$(\text{dis } A)(T)_n = A(T).$$

The category of dendroidal spaces is naturally tensored over $\mathbf{sSet}$: if X is a dendroidal space and S is a simplicial set, we define the dendroidal space $X \boxtimes S$ as, for any $T \in \Omega$,

$$(X \boxtimes S)(T) = X(T) \times S.$$

Furthermore, each dendroidal space $X : \Omega^{\text{op}} \rightarrow \mathbf{sSet}$ defines a functor

$$X^{(-)} : \mathbf{dSet}^{\text{op}} \rightarrow \mathbf{sSet}$$

by Kan extension as the unique functor (up to isomorphism) which preserves all small limits and agrees with X on representables, i.e., $X^{\Omega[T]} = X(T)$. Hence, $X^{(-)}$ is defined by the commutativity of the diagram below left, and can be computed for any $A \in \mathbf{dSet}$ with the formula below right:

$$\begin{array}{ccc} \Omega^{\text{op}} & \xrightarrow{X} & \mathbf{sSet} \\ \Omega[-] \downarrow & \nearrow X^{(-)} & \\ \mathbf{dSet}^{\text{op}} & & \end{array} \quad X^A = \lim_{\Omega[T] \rightarrow A} X(T)$$

For a given dendroidal set A , the tensoring with $\mathbf{sSet}$ and the functor $X^{(-)}$ have the following relation:

$$\mathbf{sSet}(S, X^A) \cong \mathbf{dSpaces}((\text{dis } A) \boxtimes S, X).$$

Using the functor $X^{(-)}$, we can show a characterization of the latching and matching objects of dendroidal spaces, following the same pattern of Example 1.3. For all $T \in \Omega$:

- the latching object $L_T X$ is the union of all degenerate T -dendrices,
- and the matching object satisfies $M_T X \cong X^{\partial\Omega[T]}$, where the boundary of the T -dendrex $\partial\Omega[T]$ is defined as

$$\partial\Omega[T] = \bigcup_{\substack{\alpha: S \rightarrow T \\ \alpha \text{ face map}}} \text{Im}(\Omega[\alpha]).$$

By Theorem 1.4, there exists a Reedy model structure on dendroidal spaces. Using the previous characterization together with some other properties of $\mathbf{dSpaces}$, the next theorem follows:

Theorem 2.1 ([2, Theorem 5.2]). *The category $\mathbf{dSpaces}$ admits a simplicial combinatorial proper Reedy model structure, denoted $\mathbf{dSpaces}_R$, where a map $f : X \rightarrow Y$ is:*

- a weak equivalence if, for all $T \in \Omega$, the induced maps $X(T) \rightarrow Y(T)$ are Kan weak equivalences of $\mathbf{sSet}$,
- a fibration if, for all $T \in \Omega$, the induced map

$$X^{\Omega[T]} \rightarrow X^{\partial\Omega[T]} \times_{Y^{\partial\Omega[T]}} Y^{\Omega[T]}$$

is a Kan fibration of $\mathbf{sSet}$,

- or a cofibration if, for each simplicial degree $[n] \in \Delta$, the induced map $X(-)_n \rightarrow Y(-)_n$ is a normal monomorphism of dendroidal sets.

Therefore, a Reedy fibrant object of $\mathbf{dSpaces}_R$ is a levelwise a Kan complex, but not every dendroidal space which is a levelwise Kan complex is Reedy fibrant. Furthermore, for any pair of objects $Y, Z \in \mathbf{dSpaces}$ there exist a homotopy function complex or mapping space $\text{Map}(Y, Z)$. If A is a Reedy cofibrant dendroidal space and X is a Reedy fibrant dendroidal space, the mapping space can be calculated using the tensoring with $\mathbf{sSet}$:

$$\text{Map}(A, X)_n = \mathbf{dSpaces}(A \boxtimes \Delta[n], X).$$

In particular, if A is a normal dendroidal set and X is Reedy fibrant, the functor $X^{(-)}$ is a model for the mapping space:

$$\text{Map}(\text{dis } A, X) = X^A.$$

REFERENCES

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